Acrobatics at the insect scale: A durable, precise, and agile micro–aerial robot

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Abstract

Aerial insects are exceptionally agile and precise owing to their small size and fast neuromotor control. They perform impressive acrobatic maneuvers when evading predators, recovering from wind gust, or landing on moving objects. Flapping-wing propulsion is advantageous for flight agility because it can generate large changes in instantaneous forces and torques. During flapping-wing flight, wings, hinges, and tendons of pterygote insects endure large deformation and high stress hundreds of times each second, highlighting the outstanding flexibility and fatigue resistance of biological structures and materials. In comparison, engineered materials and microscale structures in subgram micro-aerial vehicles (MAVs) exhibit substantially shorter lifespans. Consequently, most subgram MAVs are limited to hovering for less than 10 seconds or following simple trajectories at slow speeds. Here, we developed a 750-milligram flapping-wing MAV that demonstrated substantially improved lifespan, speed, accuracy, and agility. With transmission and hinge designs that reduced off-axis torsional stress and deformation, the robot achieved a 1000-second hovering flight, two orders of magnitude longer than existing subgram MAVs. This robot also performed complex flight trajectories with under 1-centimeter root mean square error and more than 30 centimeters per second average speed. With a lift-to-weight ratio of 2.2 and a maximum ascending speed of 100 centimeters per second, this robot demonstrated double body flips at a rotational rate exceeding that of the fastest aerial insects and larger MAVs. These results highlight insect-like flight endurance, precision, and agility in an at-scale MAV, opening opportunities for future research on sensing and power autonomy.